

# Kislay Tinker

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## Summary

Information Technology undergraduate (CGPA: 9.51/10) with strong foundations in Data Structures, Algorithms, Operating Systems, Computer Networks, and Software Engineering. Experienced across the stack — from concurrent backend systems in Java to applied NLP/ML pipelines in Python — with a track record of quickly learning new tools and methods to ship working software. Collaborates effectively with stakeholders to translate requirements into reliable, well-engineered solutions.

## Education

**B.Tech, Information Technology** — Arya College of Engineering, Jaipur *Expected July 2027*  
CGPA: **9.51/10** Coursework: Data Structures & Algorithms, Operating Systems, Software Engineering, DB Systems, Machine Learning

## Experience

**Data Analytics Intern** — Coderbot Robotech And IT Pvt. Ltd. (Remote) *2025 – Present*

- Gathered requirements from stakeholders and audited a client website's structure, content, and performance, producing two detailed technical audit reports identifying usability and data-quality issues
- Documented findings and actionable recommendations for stakeholders, translating raw site data into structured, client-ready deliverables that informed product decisions

## Technical Skills

**Languages:** Java, Python, C++, SQL  
**Core CS:** Data Structures & Algorithms, OOP, Operating Systems, Computer Networks, Database Management Systems, Software Engineering  
**Systems:** Multi-threading, Concurrency, Synchronization, TCP Socket Programming, Write-Ahead Logging, Client-Server Architecture  
**Tools:** Git, GitHub, MySQL, Linux, VS Code, Jupyter  
**Libraries:** Pandas, NumPy, Scikit-learn

## Projects

**Distributed Key-Value Store** [github.com/KislayTinker/Distributed-Key-Value-Store](https://github.com/KislayTinker/Distributed-Key-Value-Store)  
*Java · Multi-threading · TCP Sockets · Concurrency · Write-Ahead Logging*

- Designed and implemented a persistent key-value store from scratch in Java, exposing a custom TCP socket protocol to serve concurrent client requests
- Built a multi-threaded server handling simultaneous client connections, using explicit synchronization (locks, concurrent collections) to prevent race conditions on shared state
- Implemented write-ahead logging to disk for crash recovery, ensuring data durability across server restarts
- Applied sound engineering principles across the stack — protocol design, request handling, concurrency control, and persistence — to ensure data reliability and consistent performance under concurrent load

**In-Memory Search Engine** [github.com/KislayTinker/In-Memory-Search-Engine](https://github.com/KislayTinker/In-Memory-Search-Engine)  
*Java · HashMap · Inverted Index · OOP*

- Designed a HashMap-based inverted index data structure achieving sub-millisecond keyword lookup across the indexed corpus
- Implemented TF-based relevance ranking, improving result ordering over naive frequency-only matching

**Multilingual College AI Assistant** [github.com/KislayTinker/College\\_Bot](https://github.com/KislayTinker/College_Bot)  
*Python · NLP · Intent Classification · System Design*

- Co-designed a multilingual conversational system for Smart India Hackathon, architecting the intent-classification pipeline and query-routing logic for 50+ categories
- Engineered the end-to-end query pipeline, automating routine administrative support.

**Satellite Image Classification & Enhancement** [github.com/KislayTinker/satellite-image-classification](https://github.com/KislayTinker/satellite-image-classification)  
*Python · PyTorch · OpenCV · CNN · Streamlit*

- Designed and trained a CNN from scratch (3 conv blocks, ~2.1M parameters) to classify Sentinel-2 satellite imagery into 10 land-use categories, achieving ~90% test accuracy on EuroSAT (27,000 images)
- Built and deployed an image-enhancement pipeline (histogram equalization, sharpening, bicubic upscaling) alongside an interactive Streamlit app for end-to-end upload, enhancement, and classification

## Certifications

|                                                                                                          |      |
|----------------------------------------------------------------------------------------------------------|------|
| <b>DeepLearning.AI</b> – Supervised Machine Learning: Regression and Classification (Stanford-certified) | 2026 |
| <b>NPTEL</b> – Programming in Java (IIT-certified)                                                       | 2025 |
| <b>Deloitte Australia</b> – Data Analytics Job Simulation (Forage)                                       | 2025 |
| <b>Google Cloud</b> – Prompt Design in Vertex AI                                                         | 2025 |
| <b>Tata Consultancy Services</b> – TCS Young Professional Course                                         | 2024 |

## Leadership & Extracurriculars

- Solved 400+ problems on LeetCode, with strong proficiency in Data Structures, Algorithms, and problem-solving
- Led cross-functional academic teams of 4–6 members, delivering technical projects end-to-end within semester deadlines
- Coordinated task delegation and technical execution, improving on-time delivery for team milestones